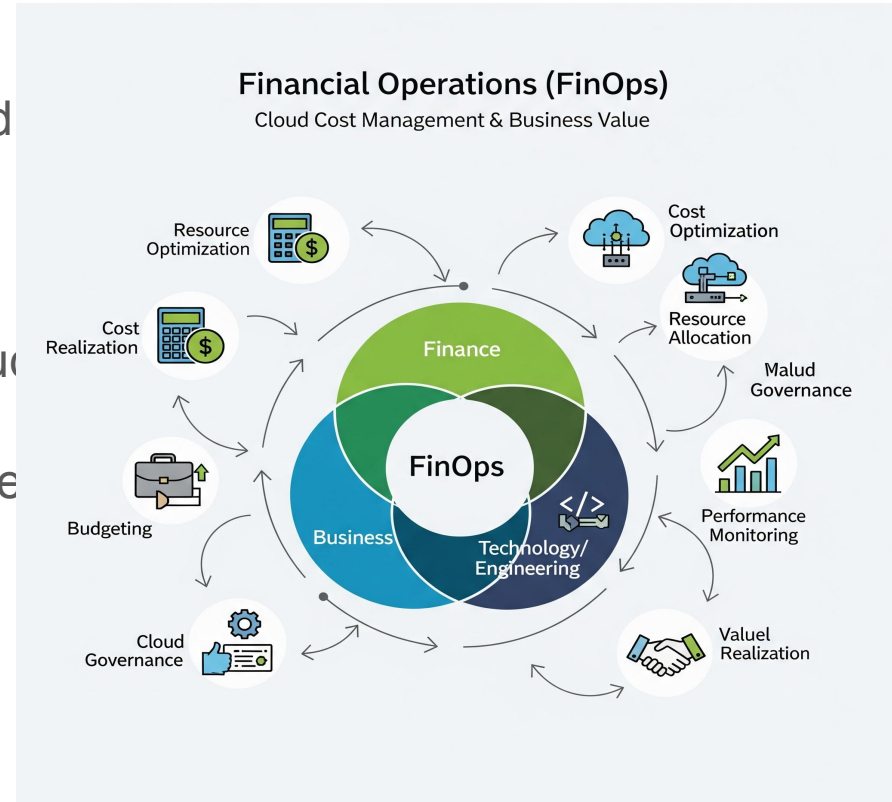


FinOps

Core Principles of FinOps

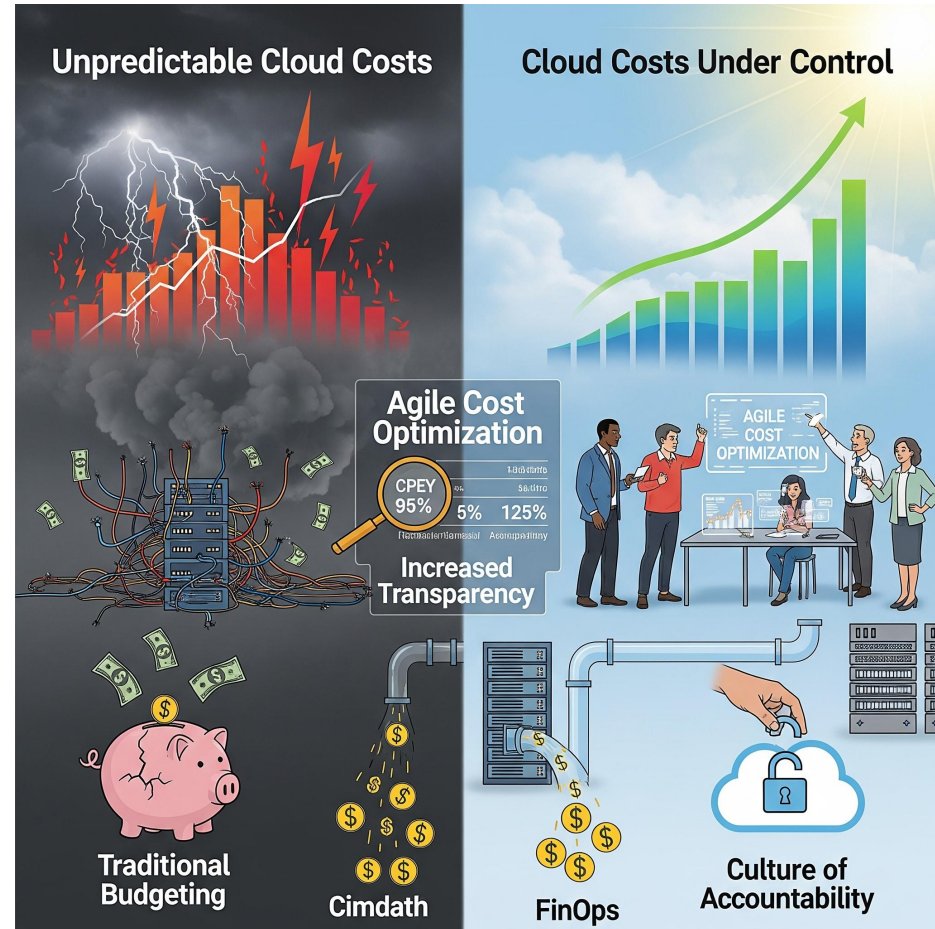
Introduction to FinOps

- ❖ Financial Operations (FinOps) = Cloud cost management discipline
- ❖ Combines financial accountability with engineering efficiency
- ❖ Goal: maximize business value of cloud investments
- ❖ Promotes collaboration across Finance, Tech, and Business
- ❖ Recognized as a cultural and operational practice



Why FinOps Matters

- ❖ Cloud costs scale unpredictably without oversight
- ❖ Traditional budgeting doesn't fit elastic cloud models
- ❖ Enables transparency of spend to decision makers
- ❖ Supports cost optimization with agility
- ❖ Builds culture of accountability



Core FinOps Principles (1)

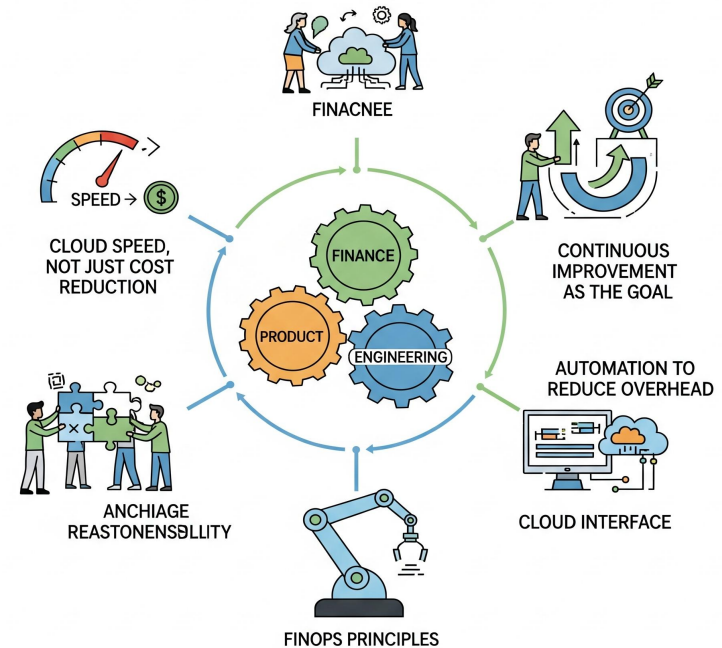
- ❖ Teams must collaborate on cloud cost decisions
- ❖ Everyone takes ownership of their cloud usage
- ❖ Decisions should be driven by business value
- ❖ Data is accessible and timely for stakeholders
- ❖ Cloud cost is variable, not fixed



Core FinOps Principles (2)

- ❖ Finance, Product, and Engineering must align
- ❖ Cloud should enable speed, not only cost reduction
- ❖ Shared responsibility is critical
- ❖ Automation where possible reduces overhead
- ❖ Continuous improvement is the goal

FINOPS PRINCIPLES



Challenges without FinOps

- ❖ Unclear ownership of cloud costs
- ❖ Unexpected billing surprises (bill shock)
- ❖ Overprovisioned or idle resources
- ❖ Lack of accountability across teams
- ❖ Difficulty in forecasting budgets



Business Outcome of FinOps

- ❖ Predictable cloud spend
- ❖ Higher ROI from cloud resources
- ❖ Transparency for executives
- ❖ Faster decision-making with data
- ❖ Improved collaboration across teams

FinOps Outcomes



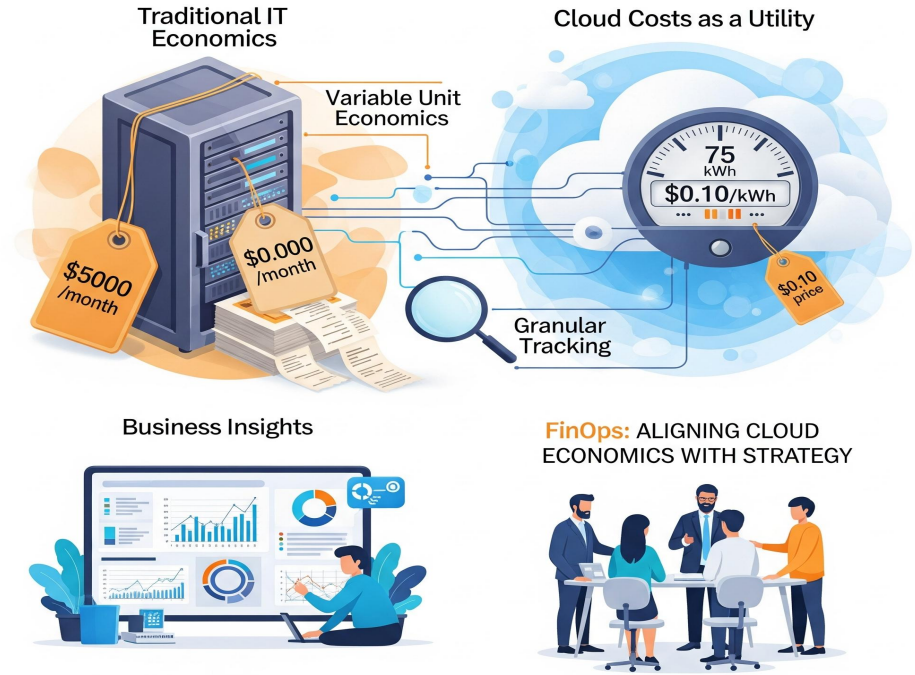
FinOps as a Cultural Practice

- ❖ More than tools: requires mindset shift
- ❖ Shared vocabulary across departments
- ❖ Finance learns technical cloud terms
- ❖ Engineers learn financial accountability
- ❖ Product learns cost-to-value trade-offs



FinOps and Cloud Economics

- ❖ Cloud costs behave like utilities (pay-per-use)
- ❖ Economics differ from traditional IT
- ❖ Understanding variable unit economics is key
- ❖ Granular tracking drives business insights
- ❖ FinOps aligns cloud economics with strategy



FinOps lifecycle at a Glance

- ❖ **Inform:** Collect and allocate cost data
- ❖ **Optimize:** Improve efficiency & automation
- ❖ **Operate:** Governance and forecasting
- ❖ Cyclical, not one-time
- ❖ Iteration improves maturity



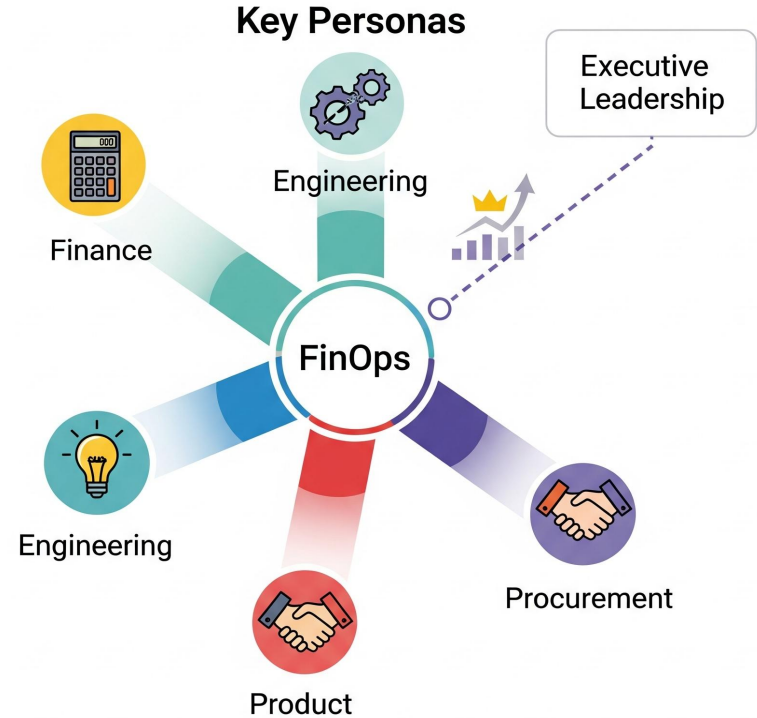
Summary of Principles

- ❖ Collaboration, ownership, and accountability
- ❖ Data-driven, value-oriented decision making
- ❖ Cloud costs are dynamic and must be managed actively
- ❖ FinOps balances cost vs speed vs quality
- ❖ Foundation for team-based cloud governance

Personas and Collaborations

Personas in FinOps

- ❖ Finance
- ❖ Engineering
- ❖ Product
- ❖ Procurement
- ❖ Executive leadership (indirect)



Finance Persona

- ❖ Focuses on budgeting and forecasting
- ❖ Needs visibility into cloud spend drivers
- ❖ Aligns costs with financial reporting
- ❖ Manages chargeback/showback models
- ❖ Ensures compliance with financial policies



Engineering Persona

- ❖ Focuses on efficient use of resources
- ❖ Responsible for rightsizing workloads
- ❖ Balances performance vs cost
- ❖ Implements automation for scaling
- ❖ Needs real-time cost data in workflows



Product Persona

- ❖ Owns product profitability and cost metrics
- ❖ Needs to balance feature velocity with spend
- ❖ Understands cost-per-customer/unit economics
- ❖ Relies on engineering + finance inputs
- ❖ Prioritizes cost efficiency as feature



Procurement Persona

- ❖ Negotiates cloud contracts with vendors
- ❖ Manages enterprise agreements and discounts
- ❖ Aligns commitments with usage trends
- ❖ Works with finance on savings plans/RIs
- ❖ Ensures vendor compliance



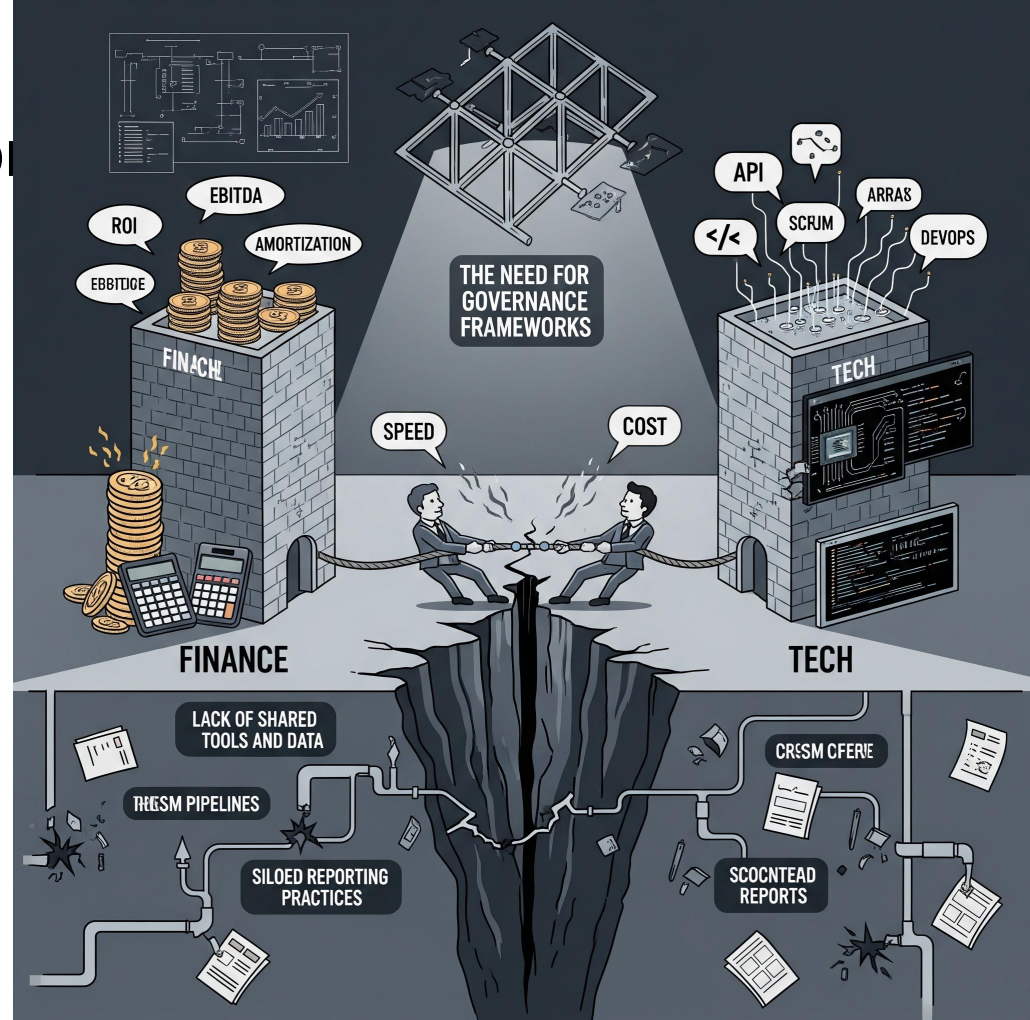
Cross-Team Collaboration

- ❖ Finance and Engineering sync on budgets vs usage
- ❖ Product drives business priorities
- ❖ Procurement secures optimal vendor deals
- ❖ Teams share dashboards & reports
- ❖ Executive sponsors reinforce accountability



Challenges in Collaboration

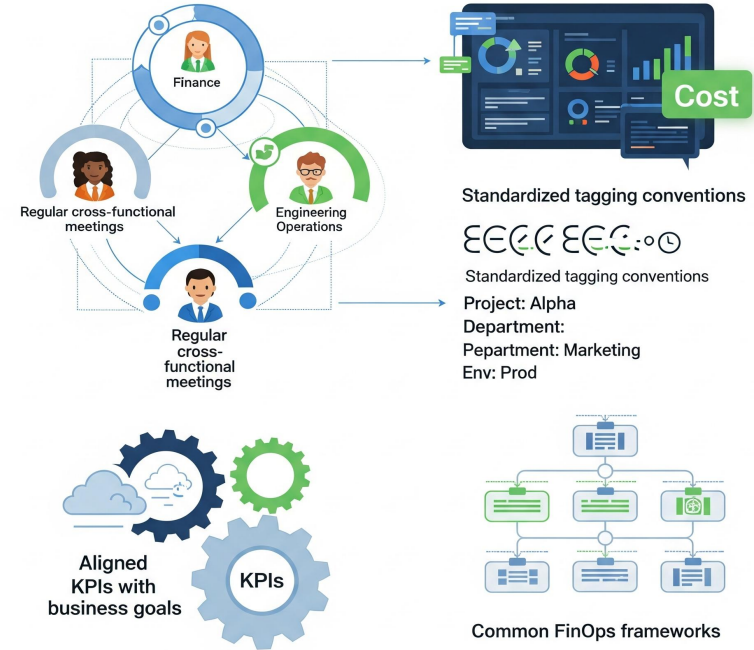
- ❖ Different vocabularies (finance vs tech)
- ❖ Conflicting priorities (speed vs cost)
- ❖ Lack of shared tools and data
- ❖ Siloed reporting practices
- ❖ Need for governance frameworks



Collaboration Best Practices

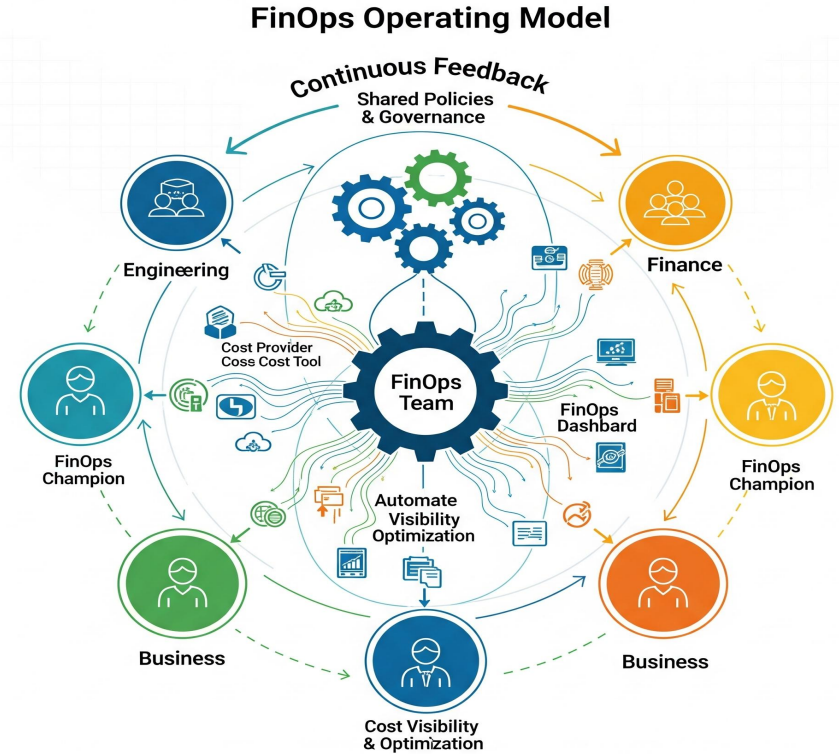
- ❖ Regular cross-functional meetings
- ❖ Shared cloud cost dashboards
- ❖ Standardized tagging conventions
- ❖ Align KPIs with business goals
- ❖ Use common FinOps frameworks

FinOps



FinOps Operating Model

- ❖ Centralized FinOps team or practice
- ❖ Embedded champions in each persona group
- ❖ Use of automated tools for cost visibility
- ❖ Governance through shared policies
- ❖ Continuous feedback loops



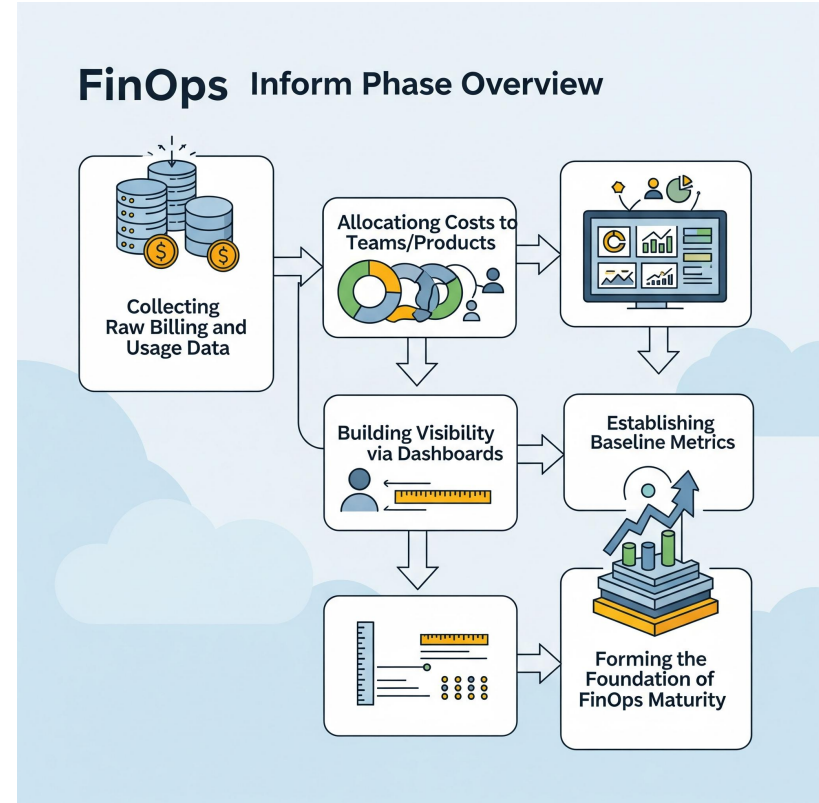
Summary – Personas & Collaboration

- ❖ Each persona brings unique expertise
- ❖ Collaboration unlocks true FinOps value
- ❖ Shared data + governance build trust
- ❖ Cross-functional accountability prevents overspend
- ❖ Personas must align around business outcomes

Inform-Optimize-Operate

Inform Phase Overview

- ❖ Collect raw billing and usage data
- ❖ Allocate costs to teams/products
- ❖ Build visibility via dashboards
- ❖ Establish baseline metrics
- ❖ Foundation of FinOps maturity

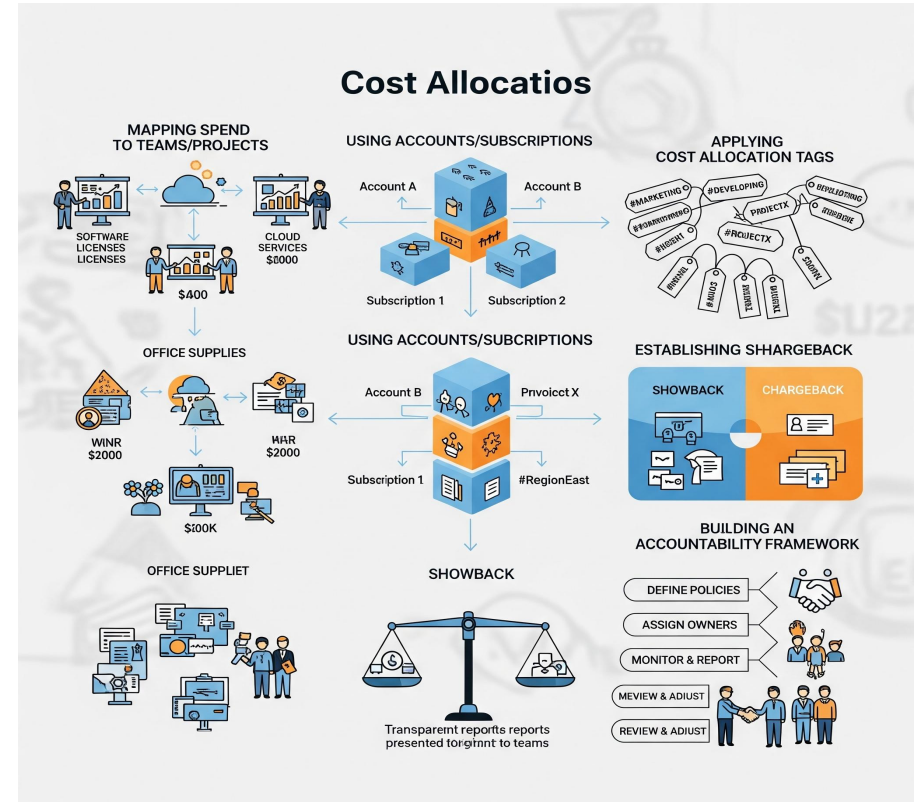


Inform-Data Sources

- ❖ AWS Cost & Usage Report (CUR)
- ❖ Azure Cost Management APIs
- ❖ GCP Billing Export to BigQuery
- ❖ Cloud provider native dashboards
- ❖ Third-party FinOps platforms

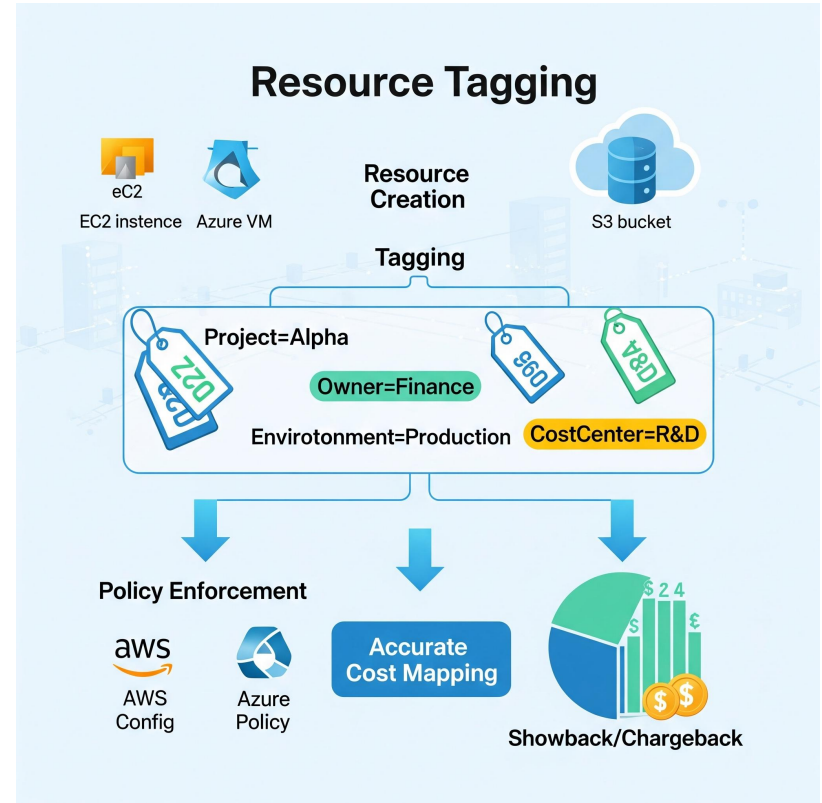


- ❖ Map spend to teams/projects
- ❖ Use accounts, subscriptions, or org units
- ❖ Apply cost allocation tags
- ❖ Establish showback/chargeback
- ❖ Build accountability framework



Inform Phase-Tagging Strategies

- ❖ Tags = key/value metadata for resources
- ❖ Examples: **Project=Alpha**, **Owner=Finance**
- ❖ Consistent tagging = accurate cost mapping
- ❖ Enforce via policies (AWS Config, Azure Policy)
- ❖ Critical for showback/chargeback



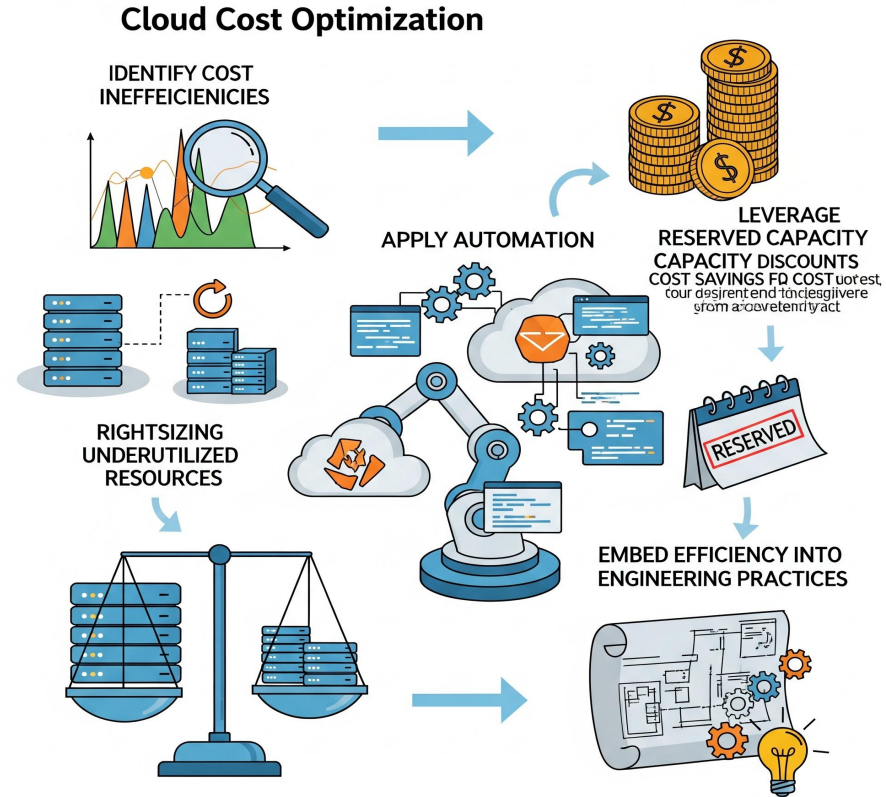
Inform-Visibility Tools

- ❖ AWS Cost Explorer
- ❖ Azure Cost Management + Power BI
- ❖ GCP Billing Reports
- ❖ Multi-cloud dashboards
- ❖ Custom Grafana/Bi integrations



Optimize Phase Overview

- ❖ Identify cost inefficiencies
- ❖ Rightsize underutilized resources
- ❖ Apply automation (scale down/up)
- ❖ Leverage reserved capacity discounts
- ❖ Embed efficiency into engineering practices



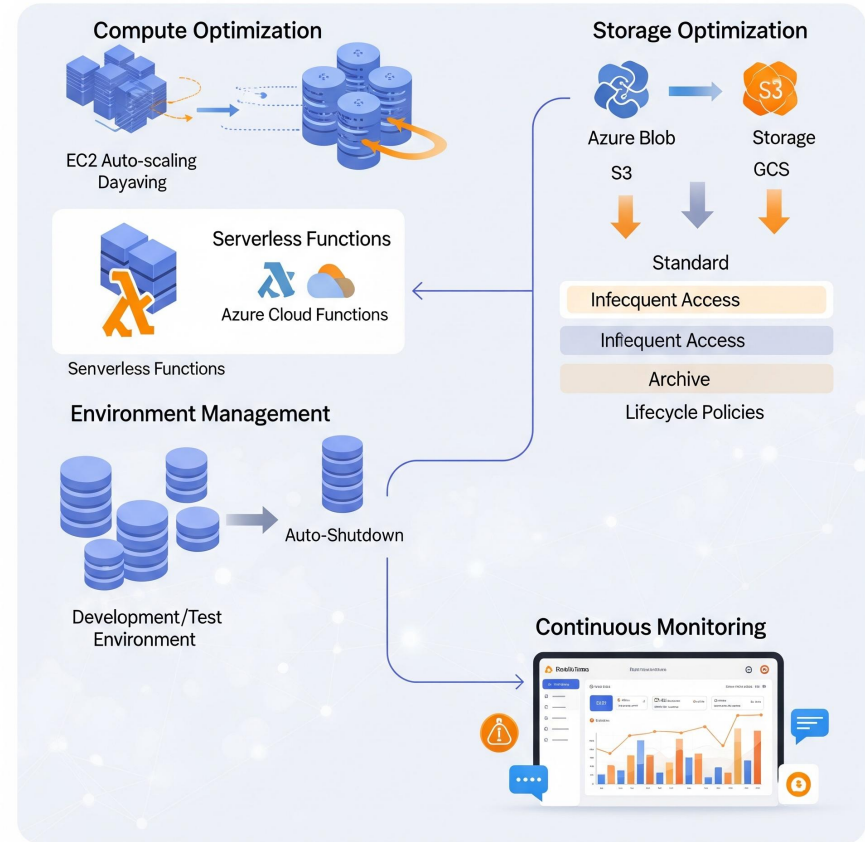
Optimize-Rightsizing

- ❖ Analyze CPU/memory metrics
- ❖ Downsize overprovisioned VMs/instances
- ❖ Scale storage tiers based on access patterns
- ❖ Terminate idle workloads
- ❖ Continuous process, not one-time



Optimize-Automation

- ❖ Auto-scaling groups for EC2
- ❖ Serverless where possible
- ❖ Auto-shutdown for dev/test
- ❖ Storage lifecycle policies (S3, Blob, GCS)
- ❖ Continuous monitoring via alerts



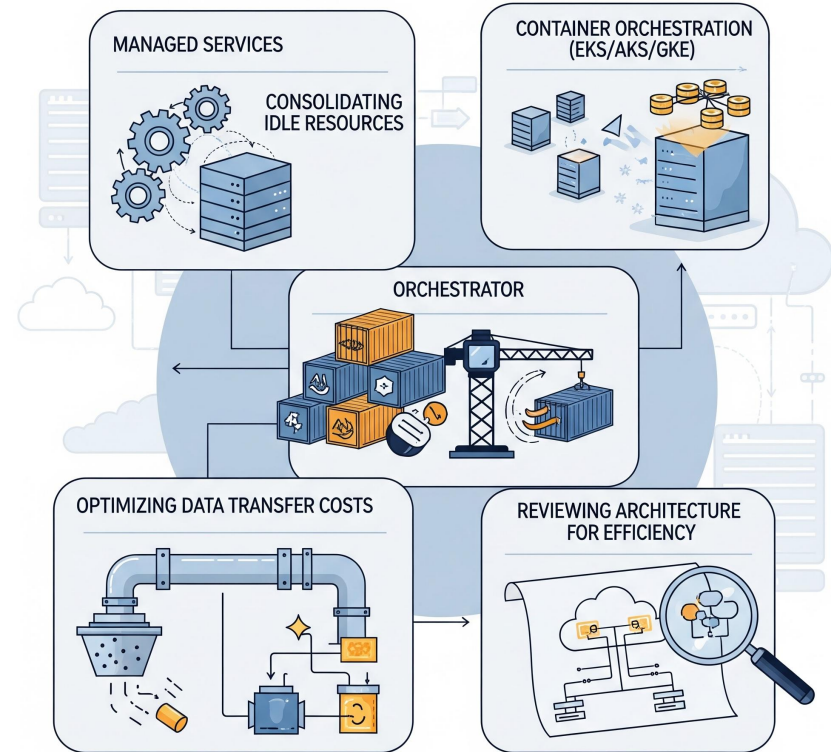
Optimize -Pricing Models

- ❖ On-demand vs Reserved Instances vs Savings Plans
- ❖ Spot instances for interruptible workloads
- ❖ Commitment-based discounts (EA, RI)
- ❖ Volume-based discounts
- ❖ Match model to workload pattern



Optimize -Efficiency Practices

- ❖ Use managed services to reduce ops cost
- ❖ Consolidate idle resources
- ❖ Container orchestration (EKS/AKS/GKE)
- ❖ Optimize data transfer costs
- ❖ Review architecture for efficiency



Operate Phase Overview

- ❖ Continuous governance of cloud costs
- ❖ Forecasting future usage/spend
- ❖ Align spend with business KPIs
- ❖ Policy-driven guardrails
- ❖ Iterative improvement over time



Operate-Governance

- ❖ Define cost guardrails (budgets/alerts)
- ❖ Enforce tagging policies
- ❖ Monitor policy compliance
- ❖ Define escalation processes
- ❖ Involve finance in governance reviews



Operate-Forecasting

- ❖ Use historical billing data
- ❖ Apply trend analysis & seasonality
- ❖ Scenario planning for new workloads
- ❖ Tie forecasts to product roadmaps
- ❖ Provide visibility to executives



Operate-Continuous Improvement

- ❖ Regular cost reviews per team
- ❖ Benchmark against past spend
- ❖ Share lessons learned across teams
- ❖ Update tagging and allocation strategies
- ❖ Refine processes every cycle



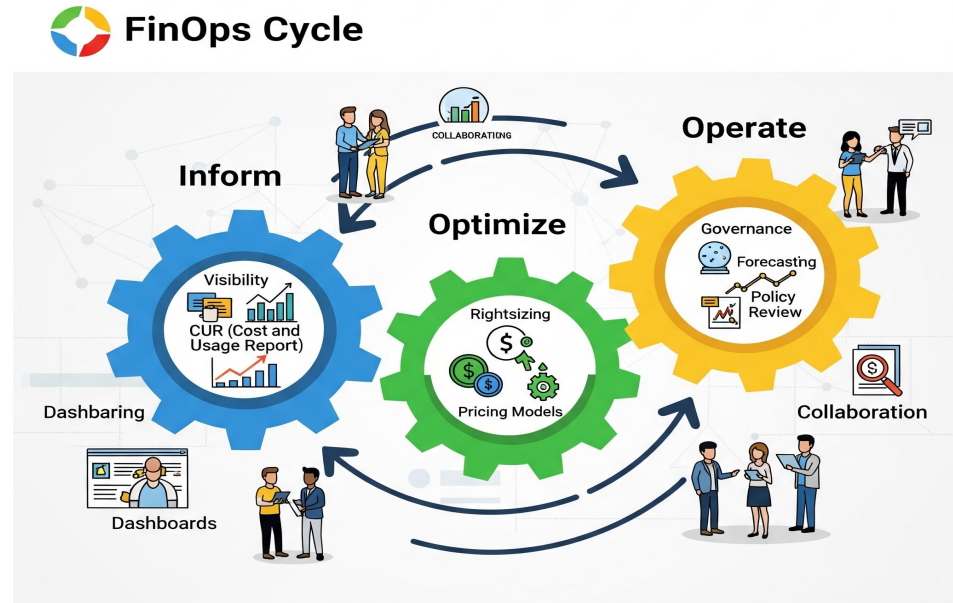
Operate-Reporting Models

- ❖ Showback: visibility without charge
- ❖ Chargeback: actual budget accountability
- ❖ Hybrid models common in enterprises
- ❖ Supports culture of ownership
- ❖ Finance ensures fairness



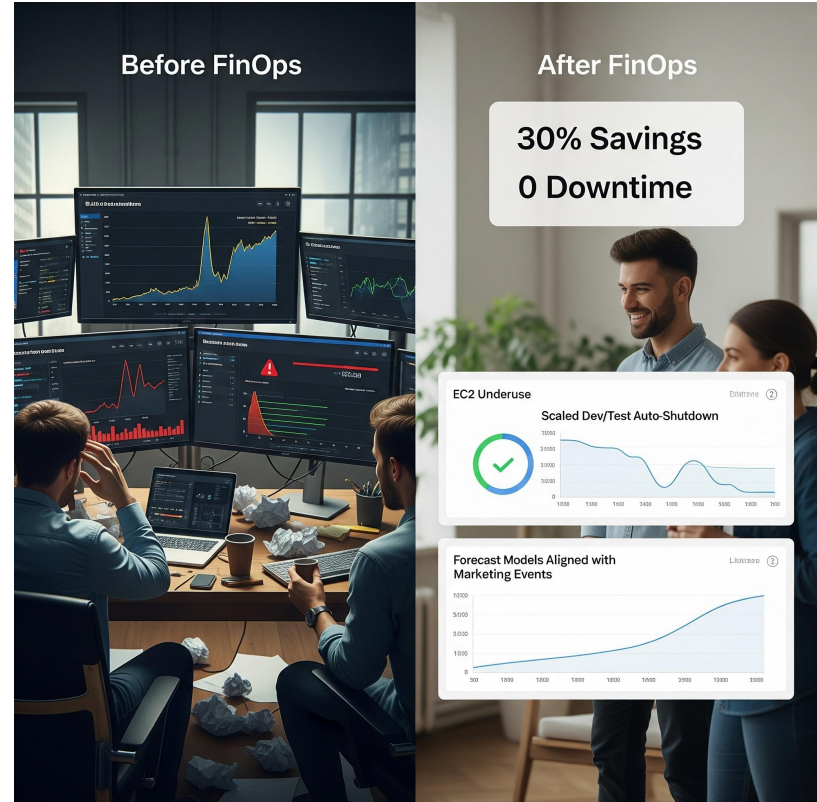
Inform-optimize-operate cycle Action

- ❖ Inform → build visibility with CUR + dashboards
- ❖ Optimize → rightsizing, automation, pricing models
- ❖ Operate → governance, forecasting, policy review
- ❖ Teams collaborate in every step
- ❖ Cycle repeats continuously



Case Study: Retail Company

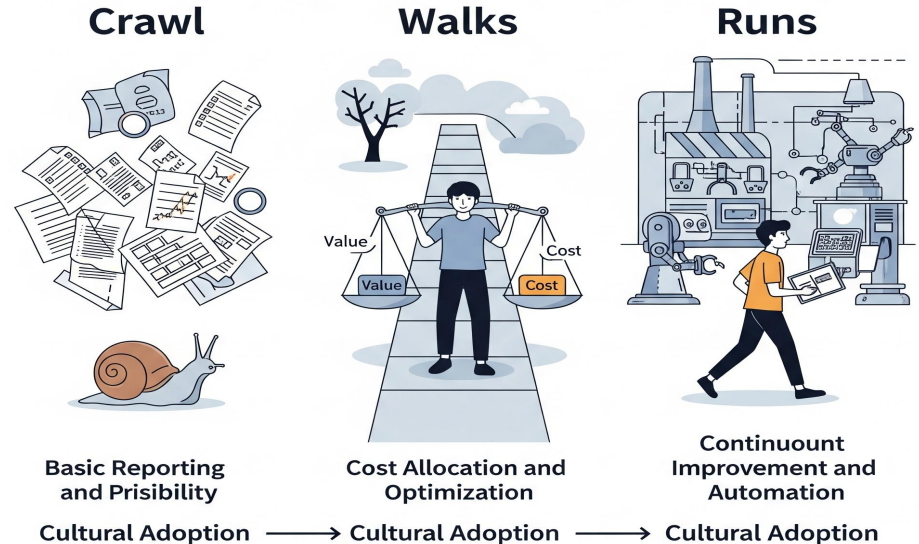
- ❖ **Problem:** unpredictable holiday season spikes
- ❖ **Inform:** dashboards showed EC2 underuse in off-peak
- ❖ **Optimize:** scaled dev/test auto-shutdown
- ❖ **Operate:** forecast models aligned with marketing events
- ❖ **Result:** 30% savings, no downtime



Maturity of FinOps Practises

- ❖ Crawl: basic reporting and visibility
- ❖ Walk: cost allocation and optimization
- ❖ Run: continuous improvement & automation
- ❖ Each phase requires cultural adoption
- ❖ Goal: maximize value, not just minimize cost
- ❖

Maturity of FinOps practices: Crawl, Walk, Run



Billing Tagging and Tools

Cloud Billing Overview

- ❖ Cloud bills are detailed, usage-based
- ❖ Contain millions of line items
- ❖ Provide granularity at resource level
- ❖ Complex but powerful for analysis
- ❖ Foundation of FinOps data



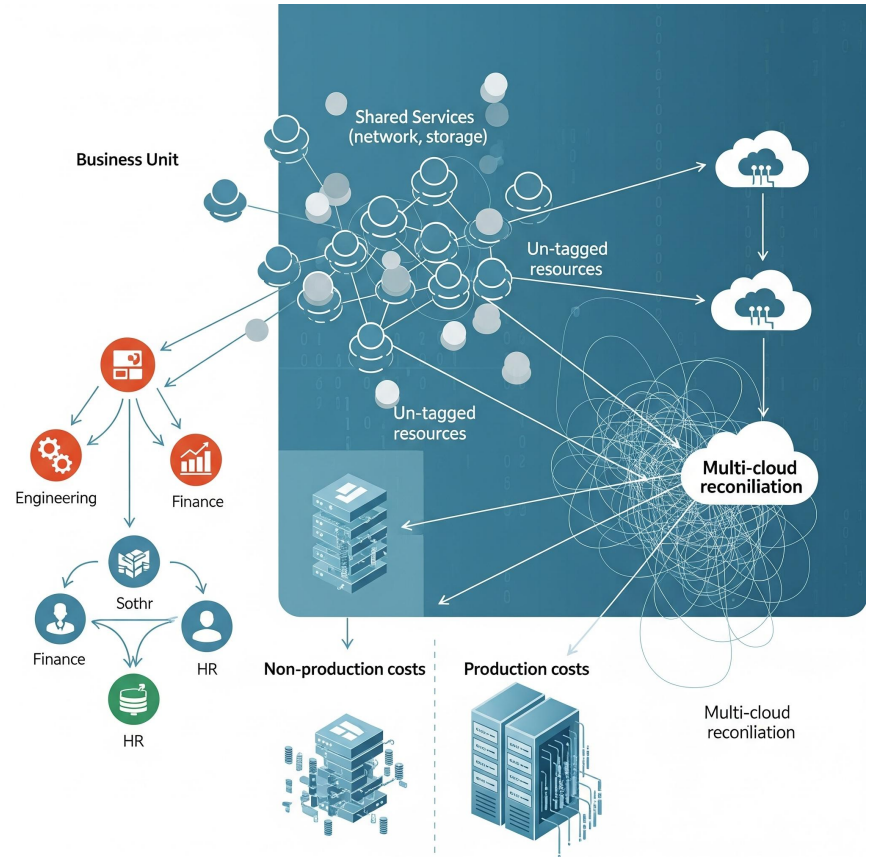
Cloud Billing Data Sources

- ❖ AWS Cost & Usage Report (CUR)
- ❖ Azure Usage + Charges API
- ❖ GCP Billing Export to BigQuery
- ❖ Third-party aggregation tools
- ❖ Linked accounts for multi-cloud



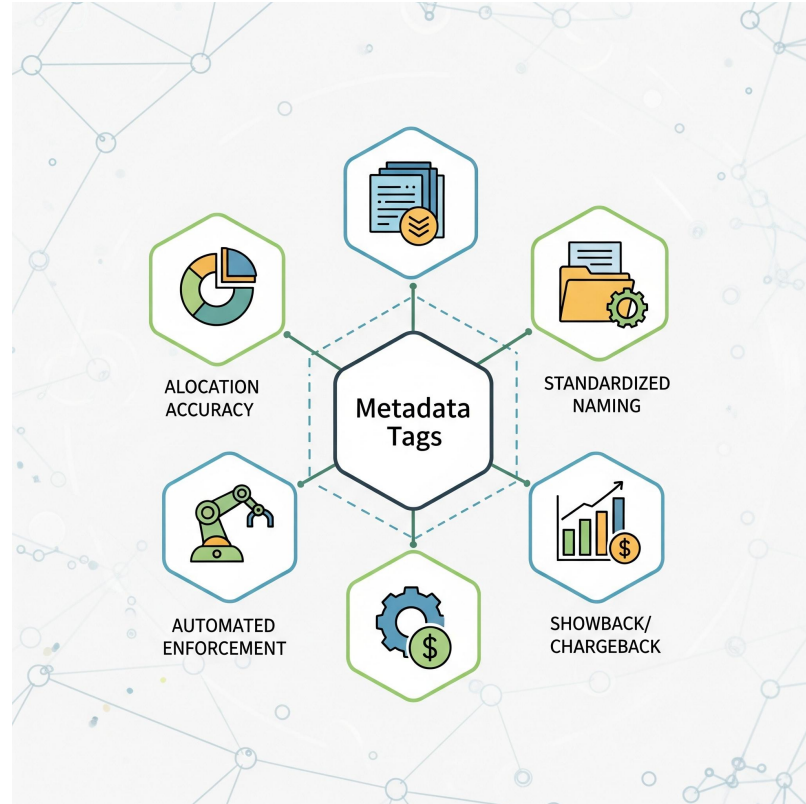
Data Allocation Challenges

- ❖ Shared services (network, storage)
- ❖ Un-tagged resources
- ❖ Non-production vs production costs
- ❖ Multi-cloud reconciliation
- ❖ Mapping to business units



Tagging Strategies Overview

- ❖ Tags = metadata for resources
- ❖ Essential for allocation accuracy
- ❖ Standardize naming conventions
- ❖ Enforce with automation
- ❖ Tags enable showback/chargeback



Tagging Best Practises

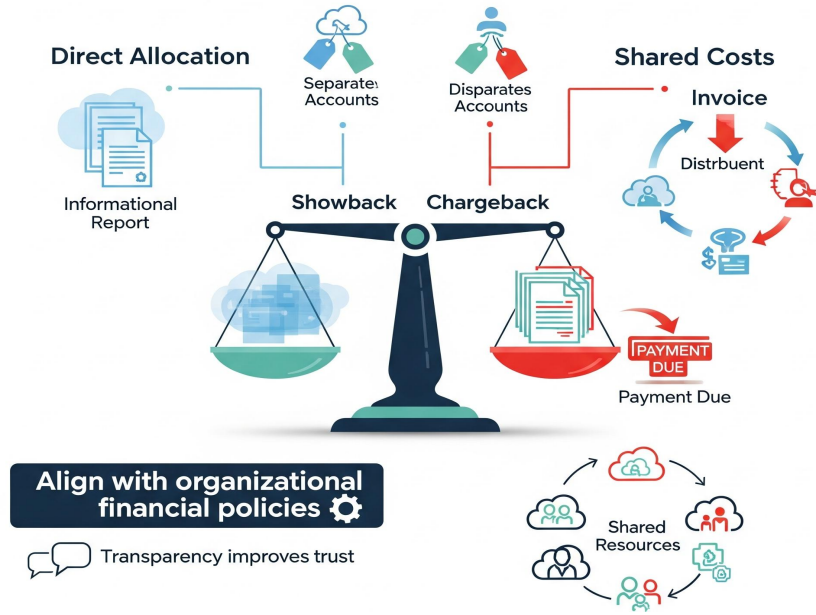
- ❖ Required tags: **Owner**, **Project**, **Environment**
- ❖ Enforce via policies or IaC templates
- ❖ Review tag coverage regularly
- ❖ Educate teams on importance
- ❖ Integrate tags into CI/CD



Cost Allocation Models

- ❖ Showback vs Chargeback
- ❖ Direct allocation via tags/accounts
- ❖ Shared costs spread proportionally
- ❖ Align with org financial policies
- ❖ Transparency improves trust

COST ALLOCATION MODELS: **SHOWBACK** vs. **CHARGEBACK**



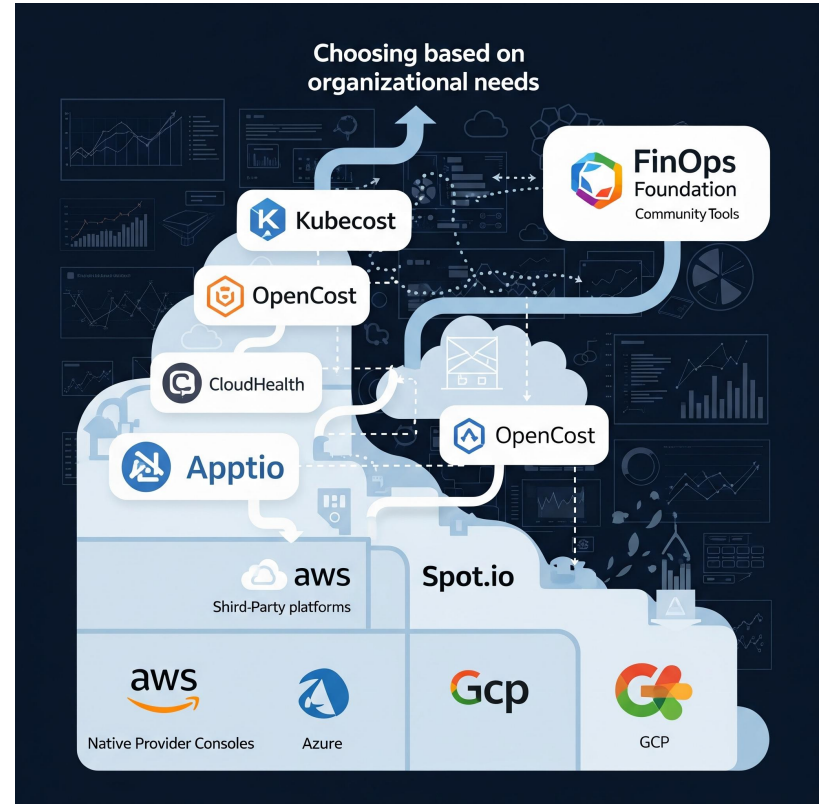
Dashboard and Visualization

- ❖ AWS Cost Explorer
- ❖ Azure Cost Management
- ❖ GCP Cloud Billing Reports
- ❖ Power BI/Tableau integration
- ❖ Custom Grafana dashboards



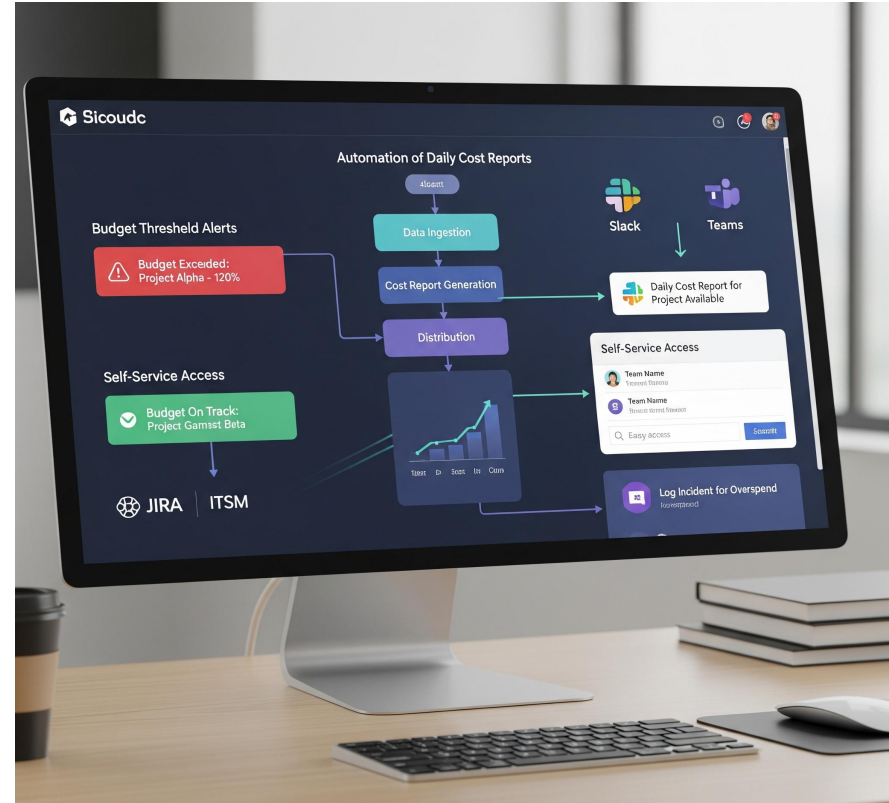
FinOps Tools Landscape

- ❖ Native: AWS, Azure, GCP consoles
- ❖ Third-party: CloudHealth, Apptio, Spot.io
- ❖ Open-source: Kubecost, OpenCost
- ❖ FinOps Foundation community tools
- ❖ Choose based on org needs



Integrating Tools into Workflow

- ❖ Automate daily cost reports
- ❖ Alerts on budget thresholds
- ❖ Integrate with Slack/MS Teams
- ❖ Provide self-service access for teams
- ❖ Tie into JIRA/ITSM for actions



Common Pitfalls in Billing &

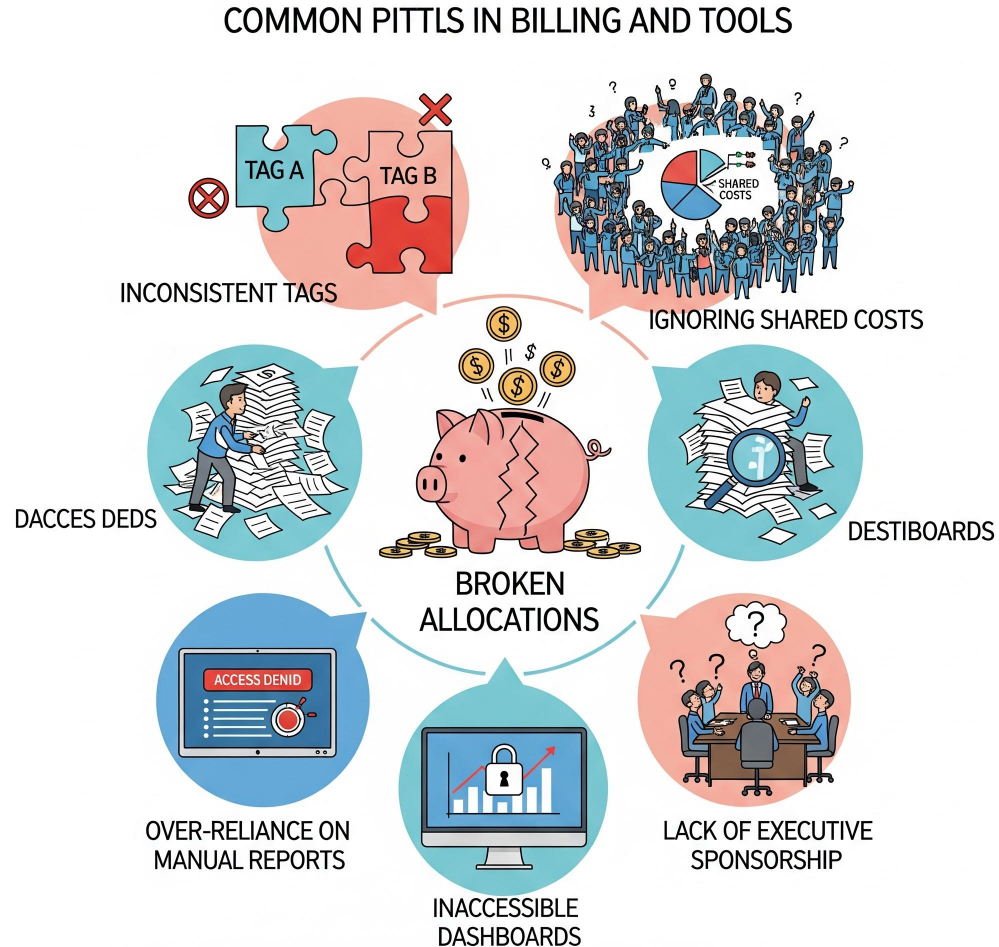
Inconsistent tags → broken allocations

Ignoring shared costs

Over-reliance on manual reports

Dashboards not accessible to all teams

Lack of executive sponsorship



Best Practices for Billing & Tools

- ❖ Automate data collection & tagging
- ❖ Provide self-service dashboards
- ❖ Review spend weekly, not monthly
- ❖ Educate all personas on reports
- ❖ Iterate tagging + dashboards continuously



Summary-Billing, Tagging and Tools

- ❖ Billing data is core to FinOps visibility
- ❖ Tagging is essential for accountability
- ❖ Allocation models drive fairness
- ❖ Dashboards/tools empower all personas
- ❖ Success = process + culture + tooling